

Personal Statement

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Throughout my life and academic experiences, I have seen the importance of *responsibility*, *independence*, and effective *leadership*. I believe these three qualities have positioned me for success in graduate school, with a goal of becoming a university professor.

I have had the opportunity to practice **responsibly** carrying out complex processes as a Discrete Mathematics TA. I managed the end-to-end duties for homework assignments: revising problems, releasing assignments, coordinating with 16 graders and 3 TAs, and publishing scores for over 600 students. To ensure that students received accurate feedback for their learning and exam preparation, I built an organizational system based on a calendar and document of weekly responsibilities. Each week, I delineated clear tasks and sent regular updates to keep our team on schedule. Even after a 25% reduction in team size, this resulted in the on-time release of all 12 homeworks last semester. Since starting this role, I have been asked to continue for the rest of my undergraduate career, and I have adapted this system to manage deadlines for ongoing research projects.

The ability to work **independently** was also essential as a TA. I created new homework questions, which required not only applying known techniques, but also walking backwards through concepts to construct relevant questions. For example, I wrote a Big-O question about proving or disproving properties related to combining time complexities. My ability to create new problems required a rigorous understanding that has proven useful in my research, such as creating code summarization problems for a human study (see other essay).

In addition, I created supplementary summer course materials for Discrete Mathematics. To encourage students to work in groups, I wanted a document that could highlight successful collaboration but also justify why collaboration is useful. I conducted a literature review to synthesize evidence into guidelines suitable for students, which has been incorporated as official course material serving 600 people each semester. To support individuals with differing incoming preparation, I also wanted a comprehensive problem set for reviewing expected material. I focused on designing questions that directly addressed common mistakes from previous semesters. I created a set of 33 practice problems for building proof-writing skills that was adopted by our institution's Focus on Fundamentals discussion section of 60 students per semester. Both of these served as opportunities for me to practice solving open-ended problems independently; for example, I wrote 100% of the collaboration document that was deployed.

Outside of my coursework, I have served in **leadership** positions in mentorship and volunteering. I tutored for the courses Elementary Programming Concepts and Applied Linear Algebra, where I used an approach of compassion and clear guidance when helping students solve questions. I also volunteered to introduce elementary school students to STEM topics, earning the highest number of volunteer hours in our chapter of IEEE-Eta Kappa Nu (EECS Honor Society). In recognition of my service and academic excellence, I received the college-wide Distinguished Leadership Award and the Arlen R. Hellwarth Award from the U-M College of Engineering.